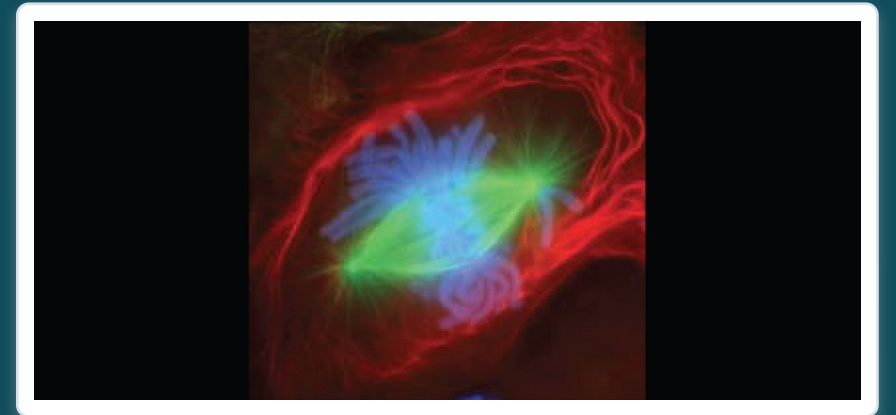


CHAPTER 3

The Cellular Level of Organization

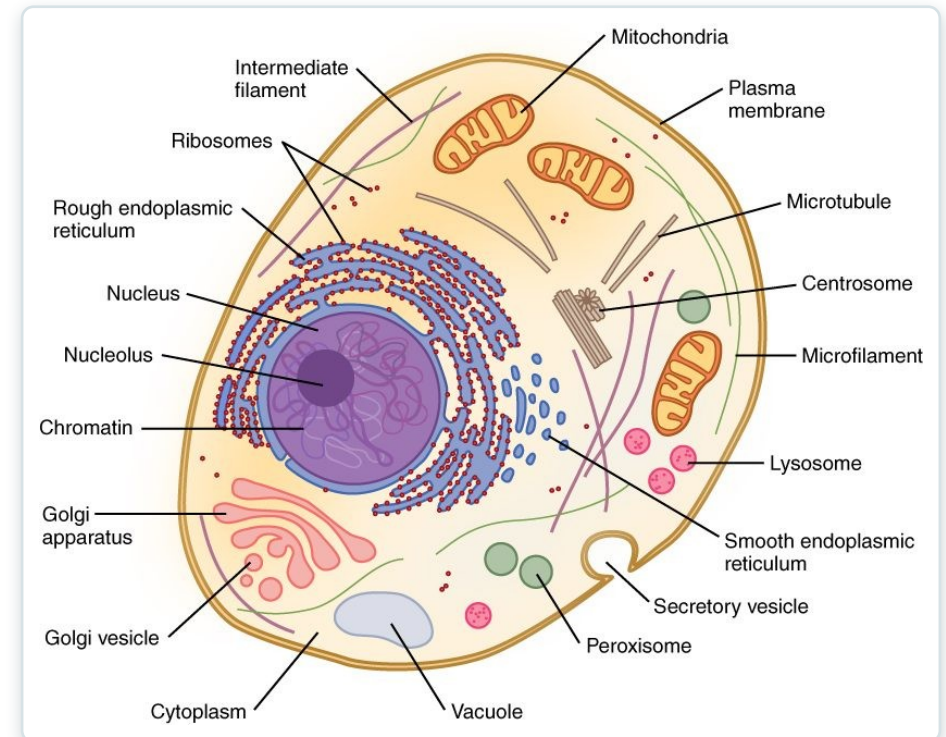
The cell membrane, organelles, DNA, and how cells divide.



Learning Objectives

After studying this chapter, students will be able to:

- 1 Describe the structure and function of the cell membrane, including its regulation of materials into and out of the cell
- 2 Describe the functions of the various cytoplasmic organelles
- 3 Explain the structure and contents of the nucleus, as well as the process of DNA replication
- 4 Explain the process by which a cell builds proteins using the DNA code
- 5 List the stages of the cell cycle in order, including the steps of cell division in somatic cells
- 6 Discuss how a cell differentiates and becomes more specialized



The cell — the basic unit of life.



3.1

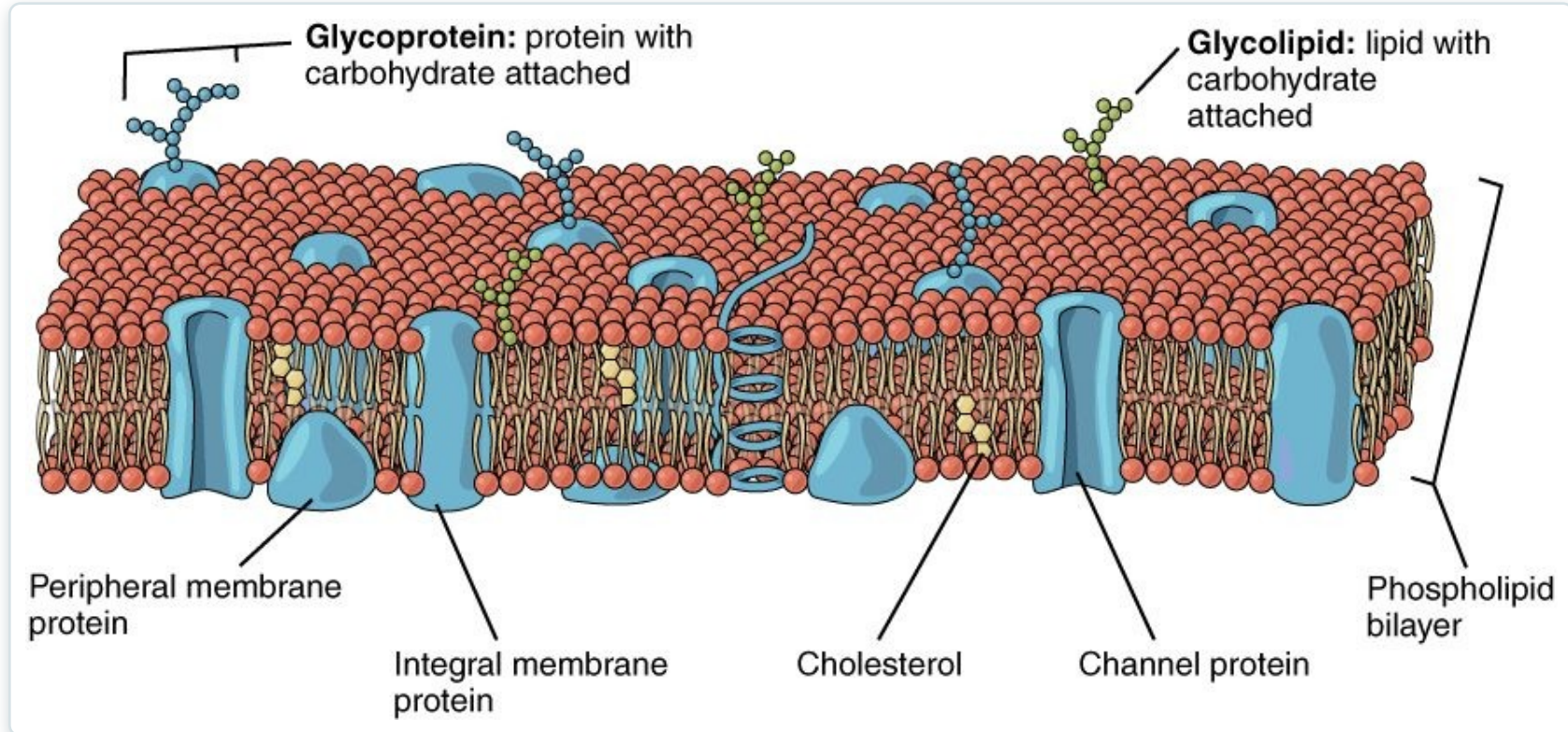
SECTION

The Cell Membrane

The selectively permeable boundary that controls what enters and leaves the cell.

SECTION 3.1

The Cell Membrane

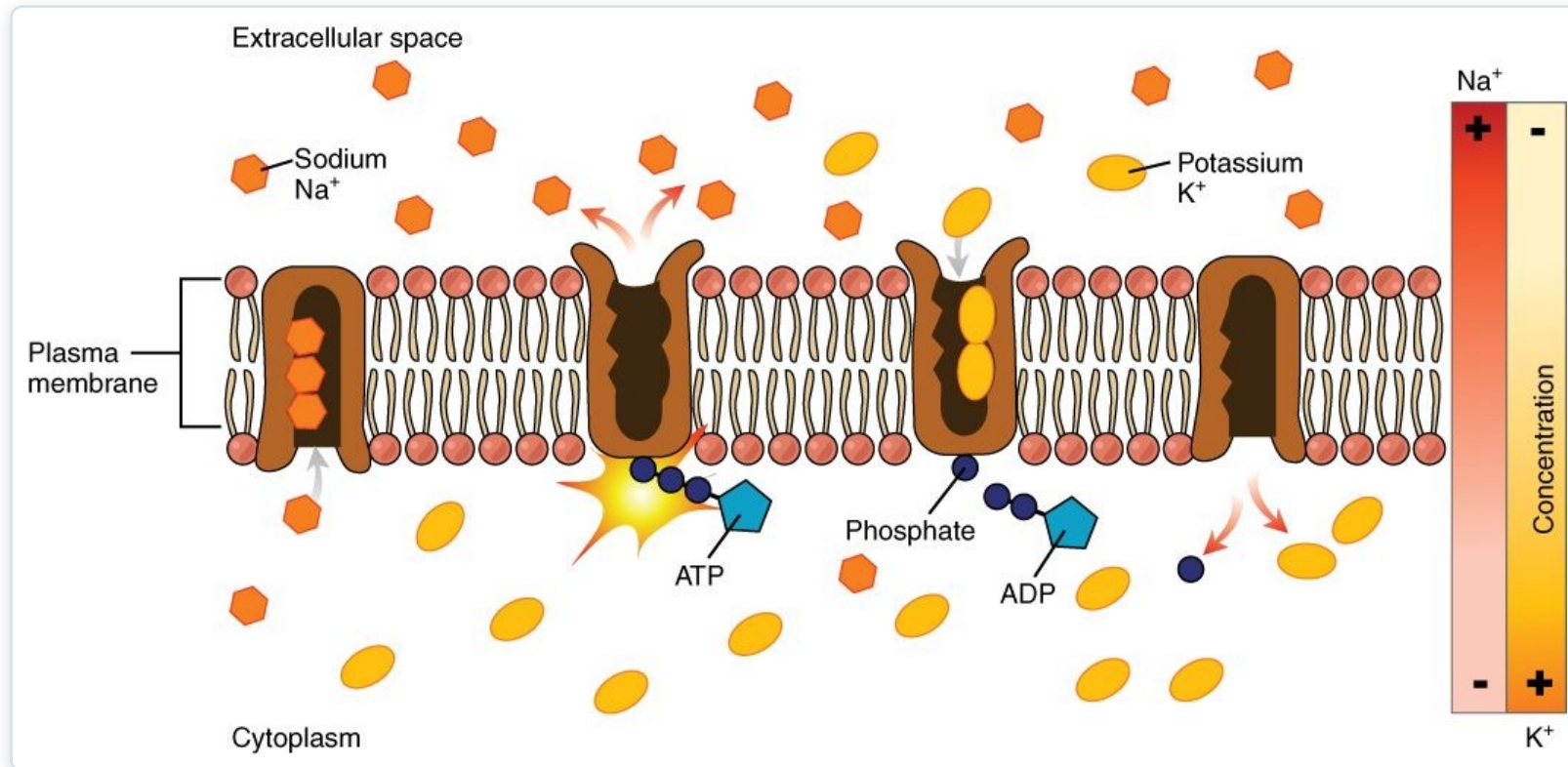


The fluid-mosaic model of the cell membrane

- Phospholipid bilayer with embedded proteins
- Hydrophilic heads out, hydrophobic tails in
- Cholesterol stabilizes the membrane
- Selectively permeable barrier

SECTION 3.1

Active Transport

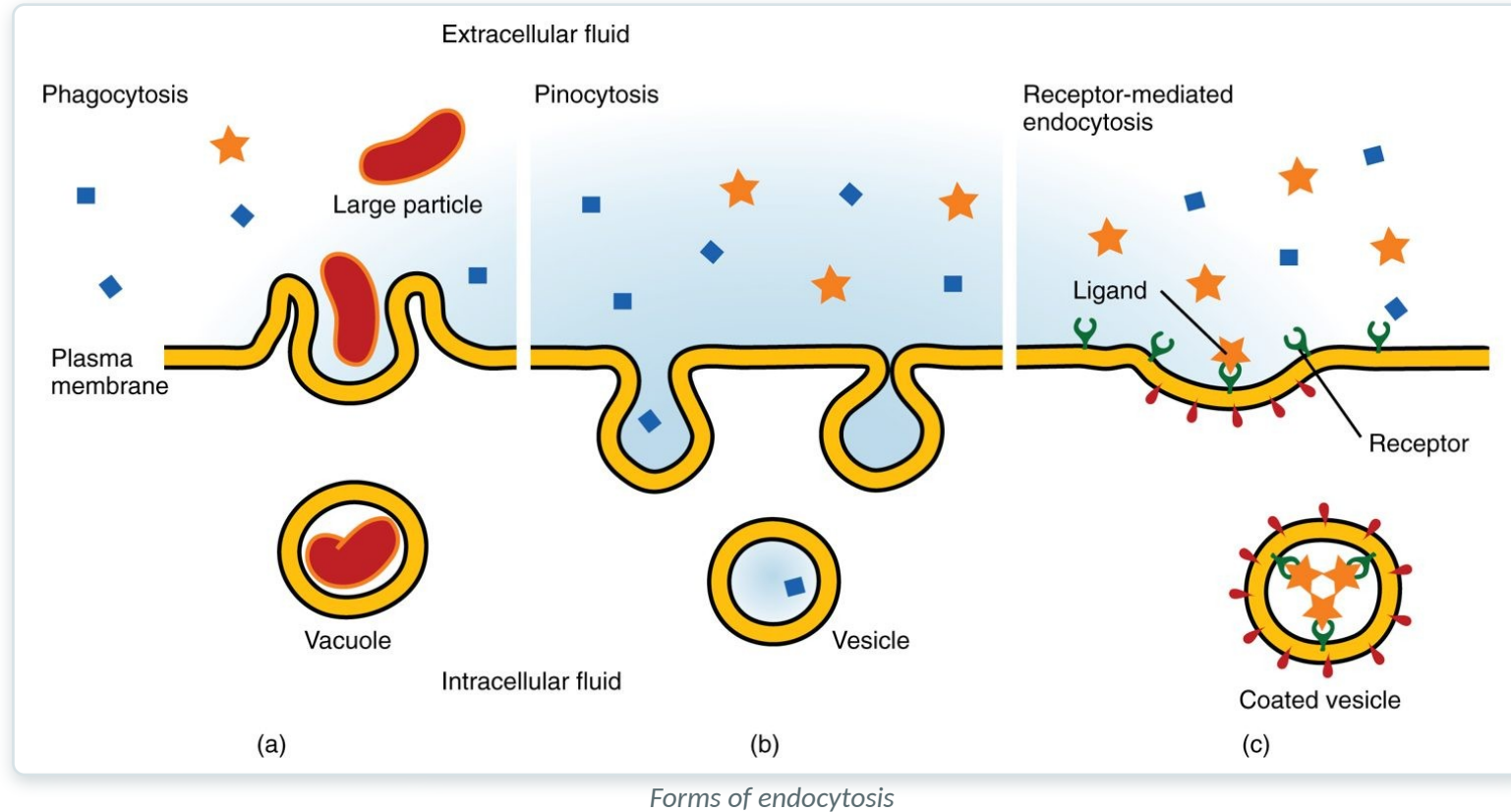


Active transport across the membrane

- Moves substances against their gradient
- Requires energy (ATP)
- Pumps maintain ion gradients
- Example: the sodium-potassium pump

SECTION 3.1

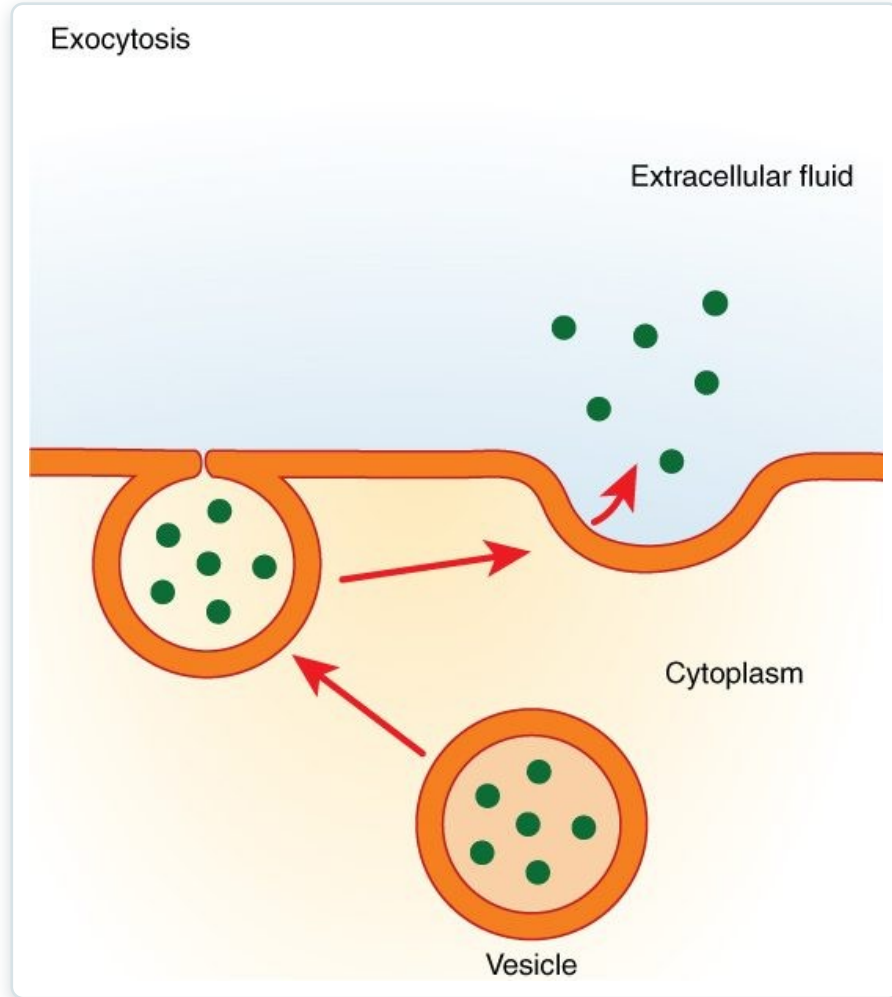
Endocytosis



- Bulk intake of material into the cell
- Phagocytosis = "cell eating"
- Pinocytosis = "cell drinking"
- Receptor-mediated for specific targets

SECTION 3.1

Exocytosis



Exocytosis releasing material from the cell

- Bulk export out of the cell
- Vesicles fuse with the membrane
- Secretes hormones, enzymes, mucus
- The reverse of endocytosis



3.2

SECTION

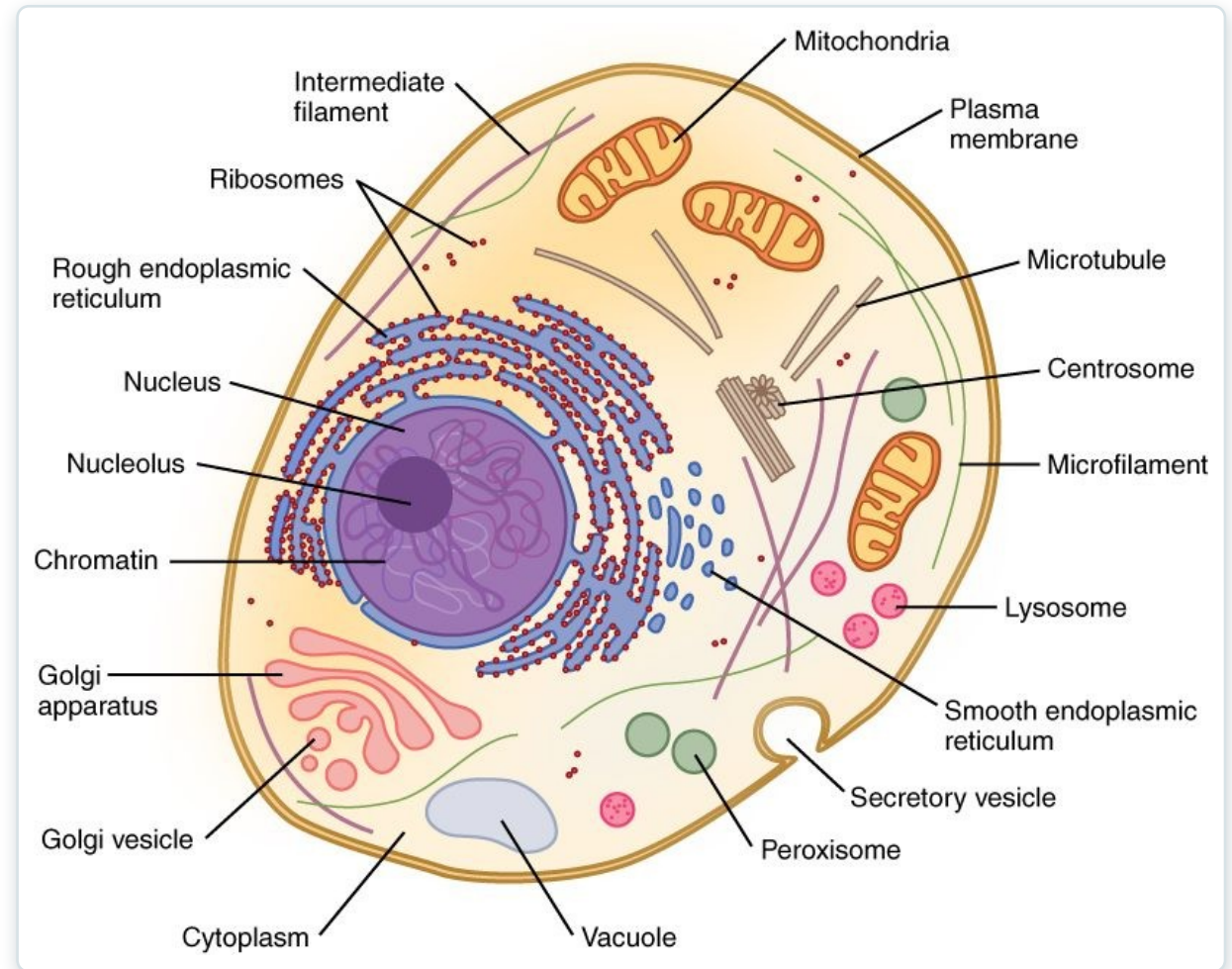
The Cytoplasm and Cellular Organelles

The cell's internal compartments and the job each one does.

SECTION 3.2

The Prototypical Cell

- Cytoplasm holds organelles in cytosol
- Each organelle has a specialized job
- Division of labor keeps the cell efficient
- The nucleus directs activities

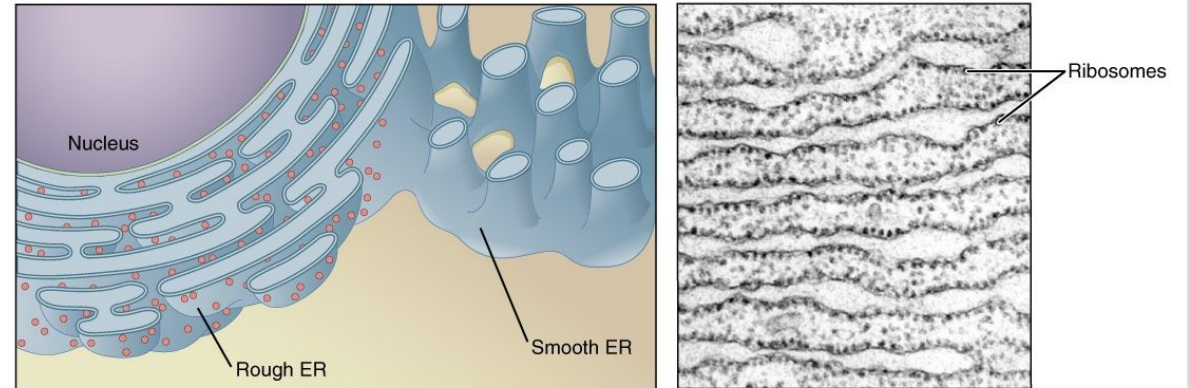


A composite (prototypical) human cell

SECTION 3.2

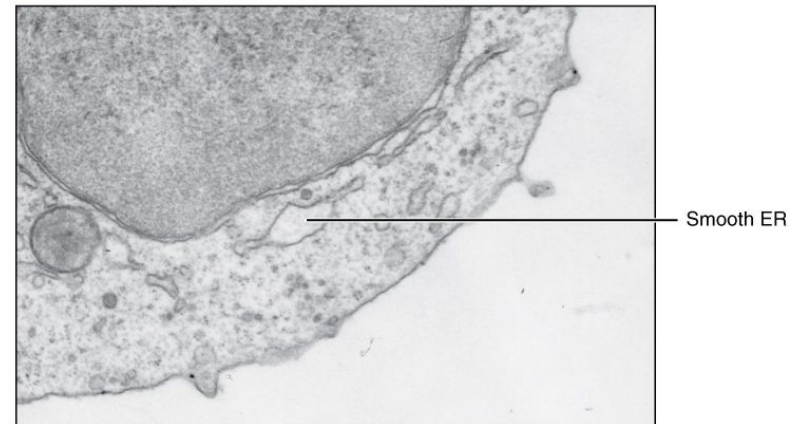
The Endoplasmic Reticulum

- Rough ER is studded with ribosomes
- Rough ER makes and folds proteins
- Smooth ER makes lipids and detoxifies
- A continuous membrane network



(a)

(b)



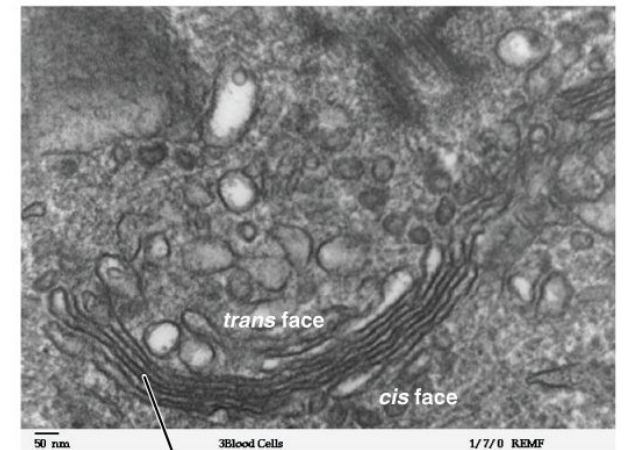
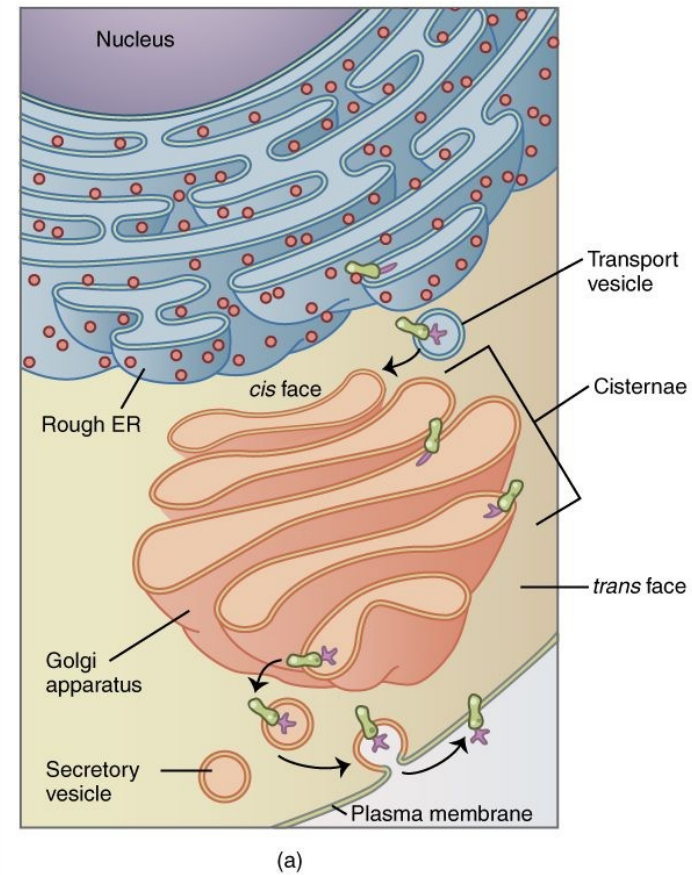
(c)

Rough and smooth endoplasmic reticulum

SECTION 3.2

The Golgi Apparatus

- Modifies, sorts, and packages proteins
- Receives vesicles from the ER
- Ships finished products in vesicles
- The cell's "post office"



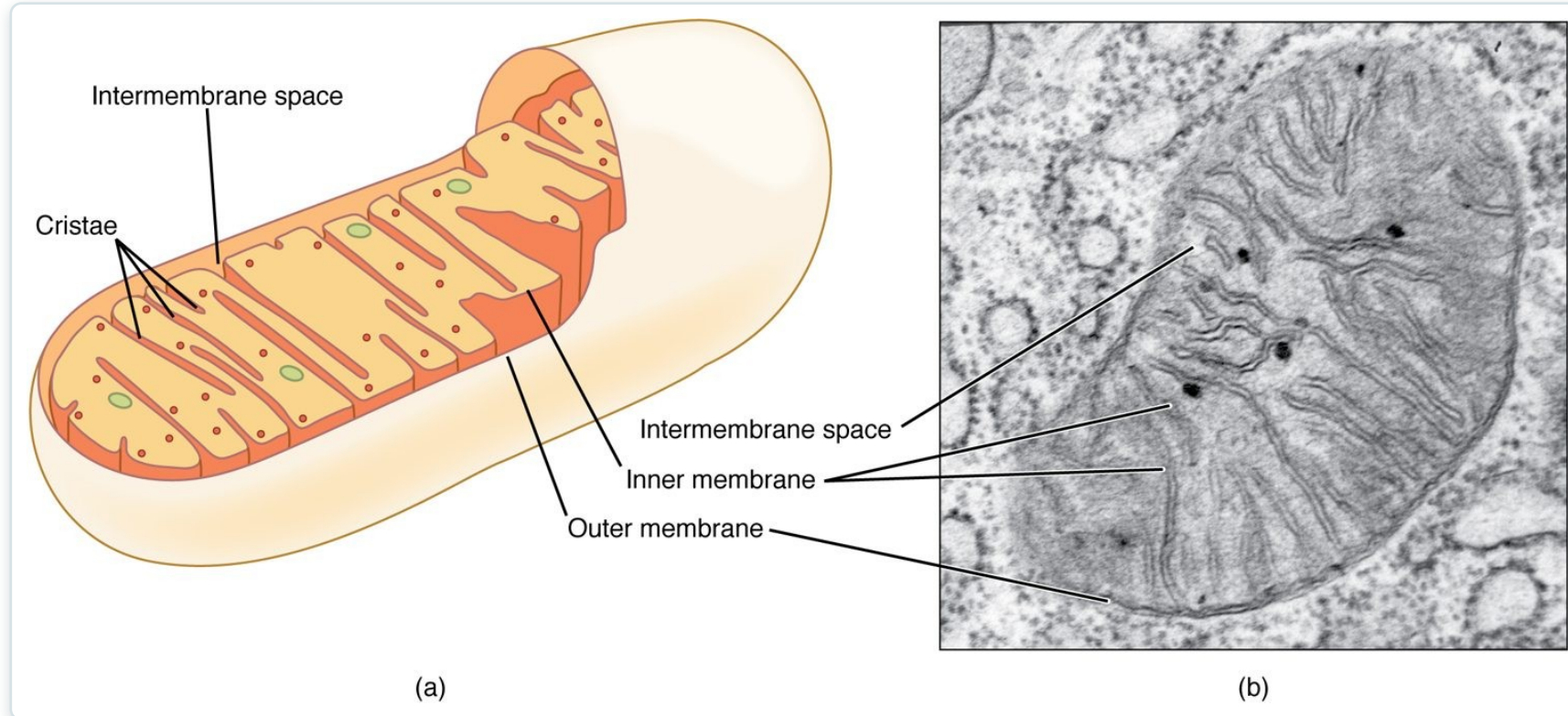
Golgi apparatus

(b)

The Golgi apparatus

SECTION 3.2

Mitochondria



Structure of a mitochondrion

- Site of cellular respiration
- Produce ATP — the cell's energy
- Folded inner membrane (cristae)
- Carry their own DNA



3.3

SECTION

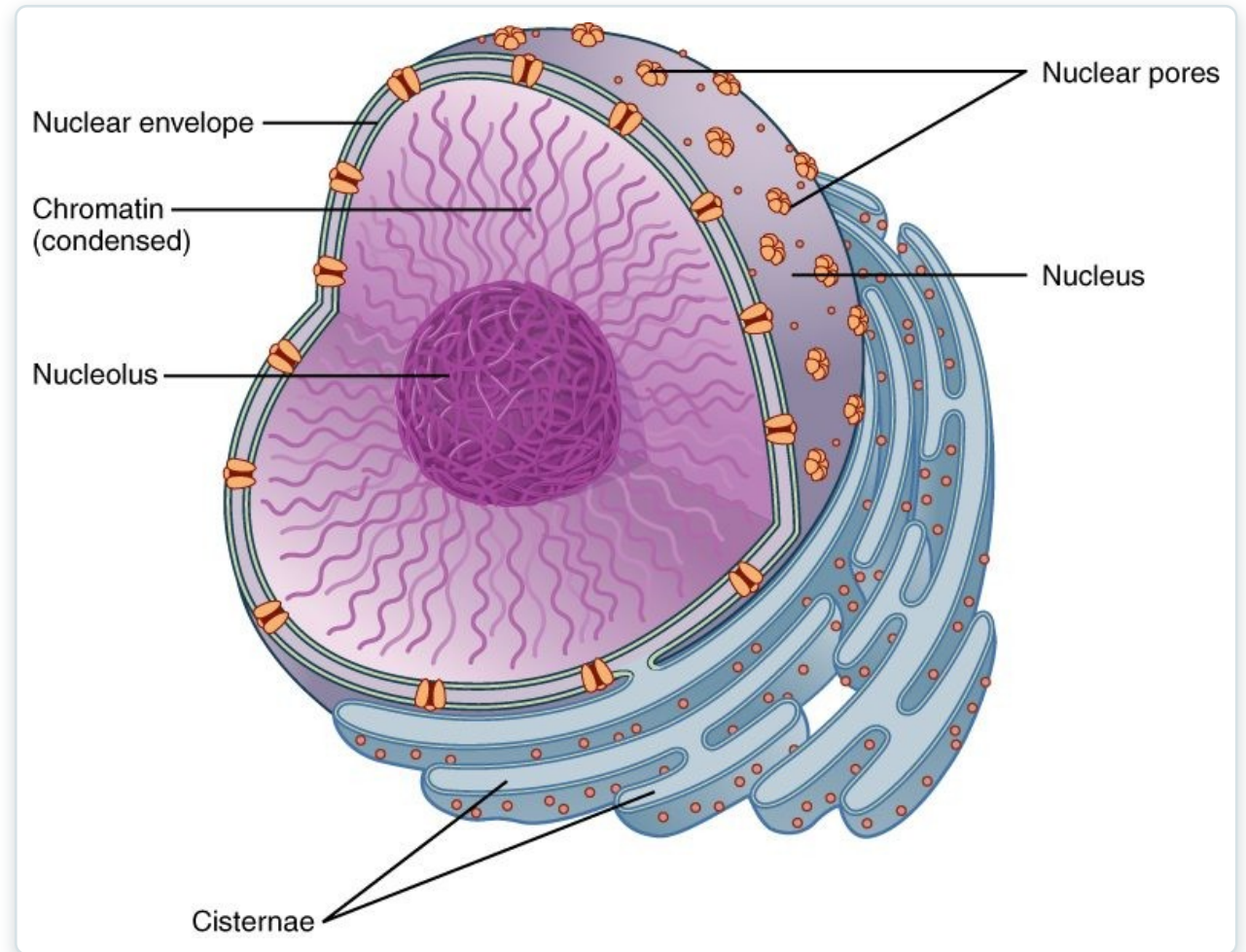
The Nucleus and DNA Replication

The control center and how the cell copies its genetic blueprint.

SECTION 3.3

The Nucleus

- Houses the cell's DNA
- Nuclear envelope with pores
- Nucleolus makes ribosomes
- Directs all cell activities

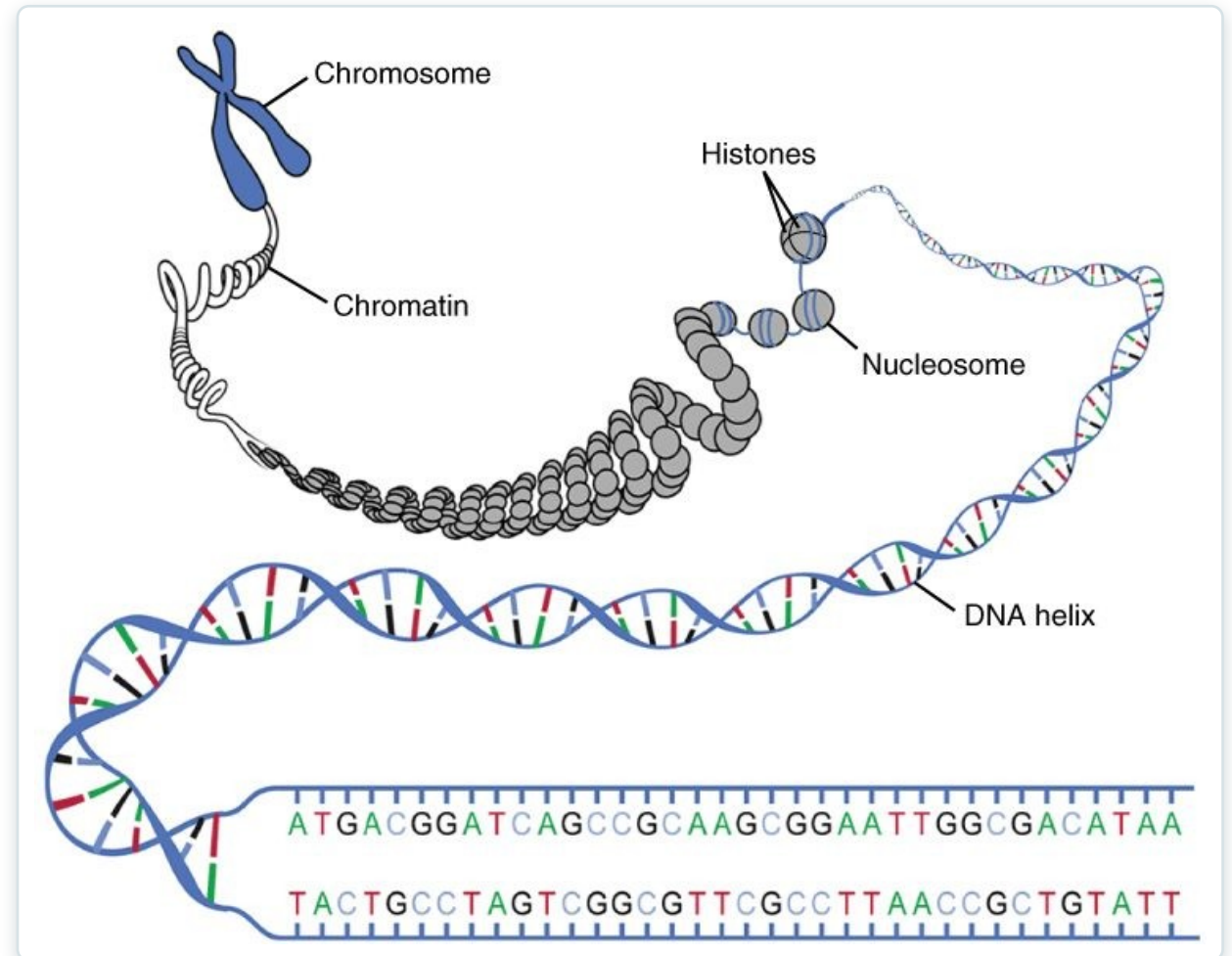


Structure of the nucleus

SECTION 3.3

DNA Packaging

- DNA wraps around histone proteins
- Forms nucleosomes, then chromatin
- Condenses into chromosomes to divide
- Packs ~2 m of DNA into each nucleus

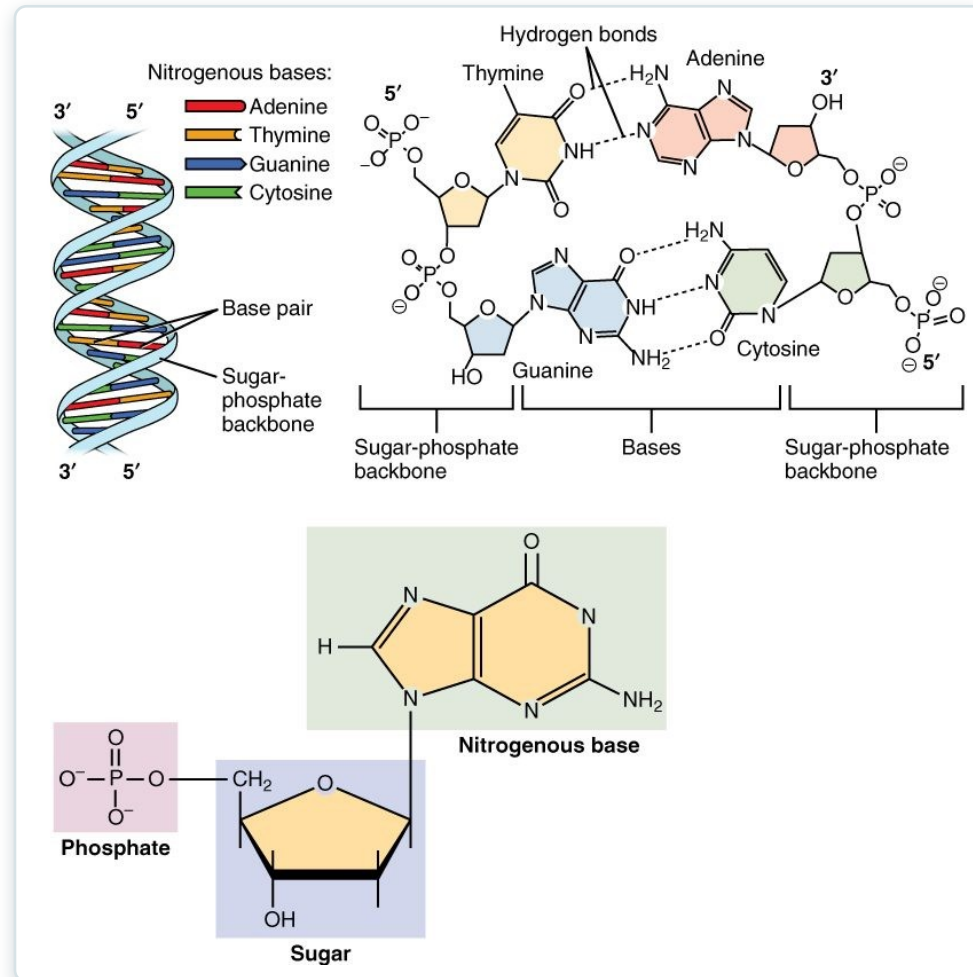


Levels of DNA packing in the chromosome

SECTION 3.3

DNA Structure

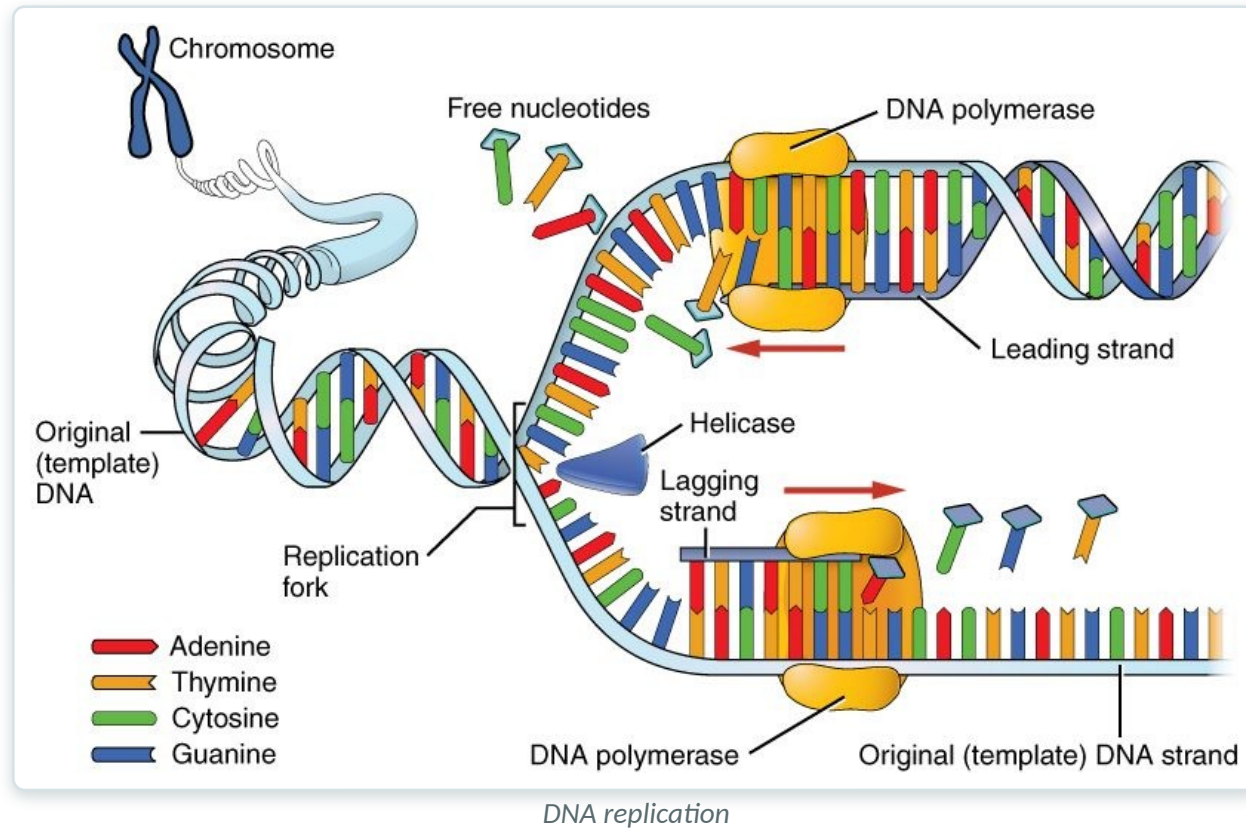
- Double helix of two strands
- Sugar-phosphate backbone
- Bases pair: A-T and G-C
- The sequence is the genetic code



The structure of DNA and base pairing

SECTION 3.3

DNA Replication



- DNA copies itself before division
- Helicase unwinds the double helix
- DNA polymerase builds new strands
- Each daughter cell gets identical DNA



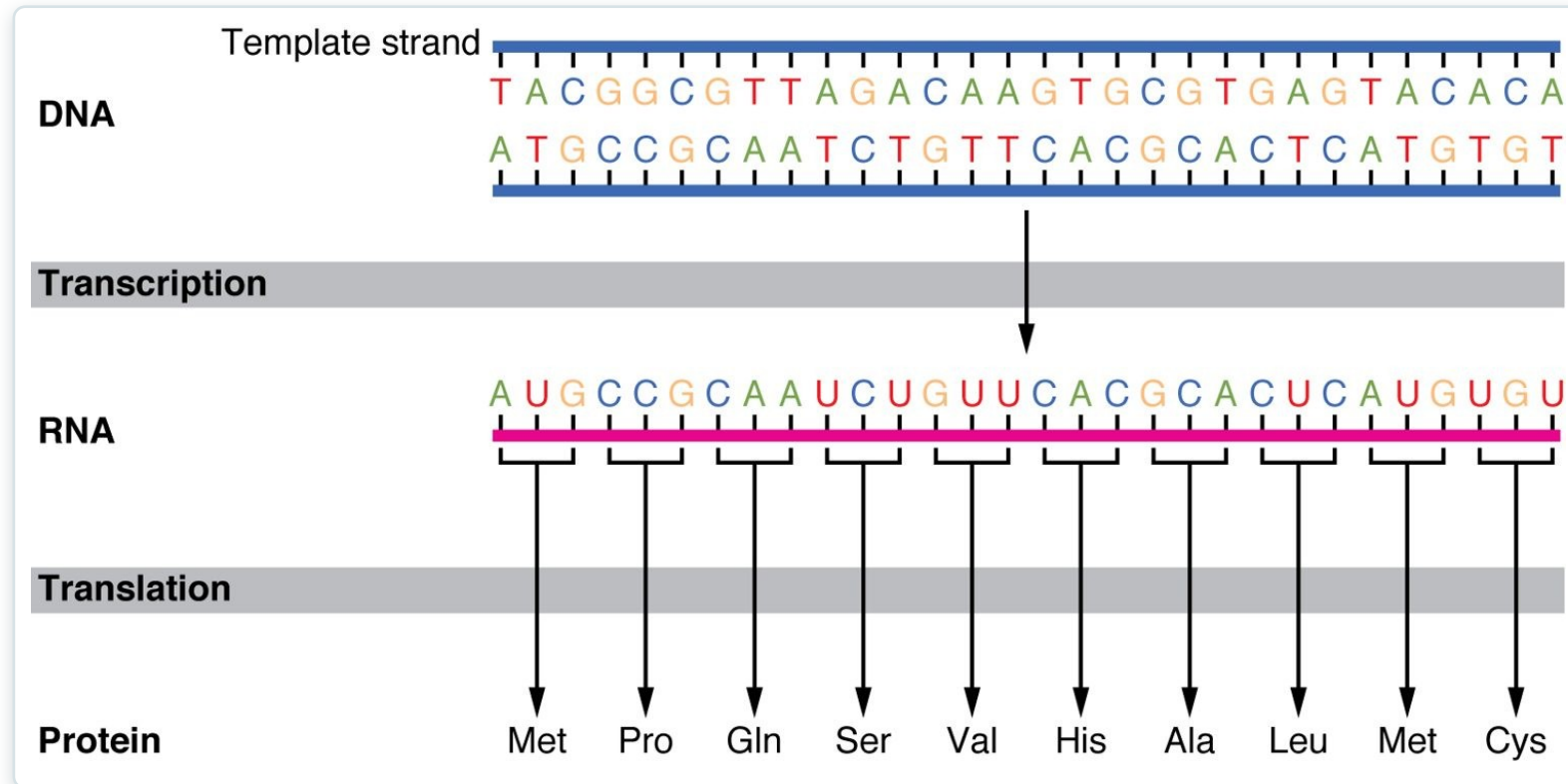
3.4

SECTION

Protein Synthesis

Reading the genetic code to build the proteins that run the cell.

From Gene to Protein

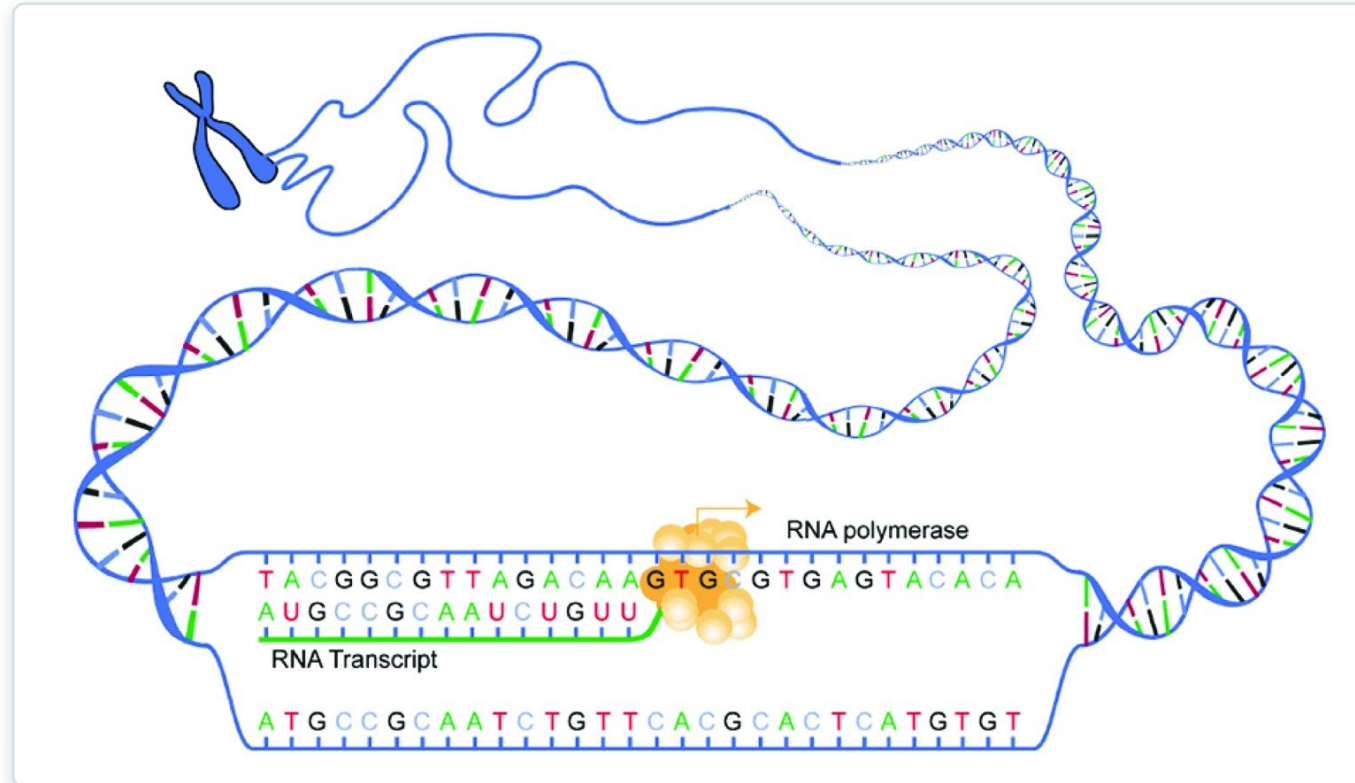


Overview of transcription and translation

- DNA → RNA → protein (central dogma)
- Transcription copies DNA into mRNA
- Translation builds protein from mRNA
- Codons specify amino acids

SECTION 3.4

Transcription

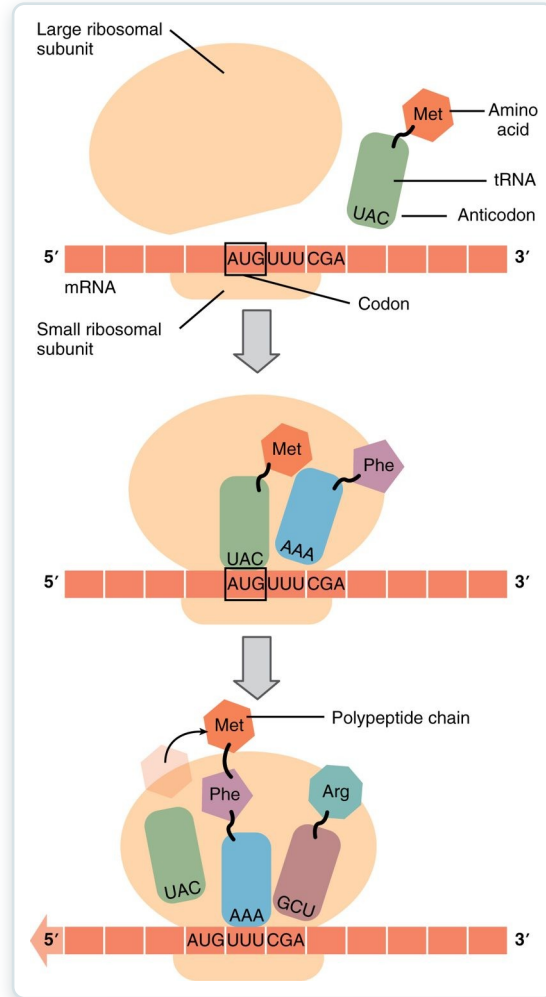


Transcription of DNA into mRNA

- RNA polymerase reads the DNA template
- Builds a complementary mRNA strand
- Occurs in the nucleus
- mRNA carries the message out

SECTION 3.4

Translation



Translation at the ribosome

- The ribosome reads mRNA codons
- tRNA delivers matching amino acids
- Amino acids link into a polypeptide
- Produces the finished protein



3.5

SECTION

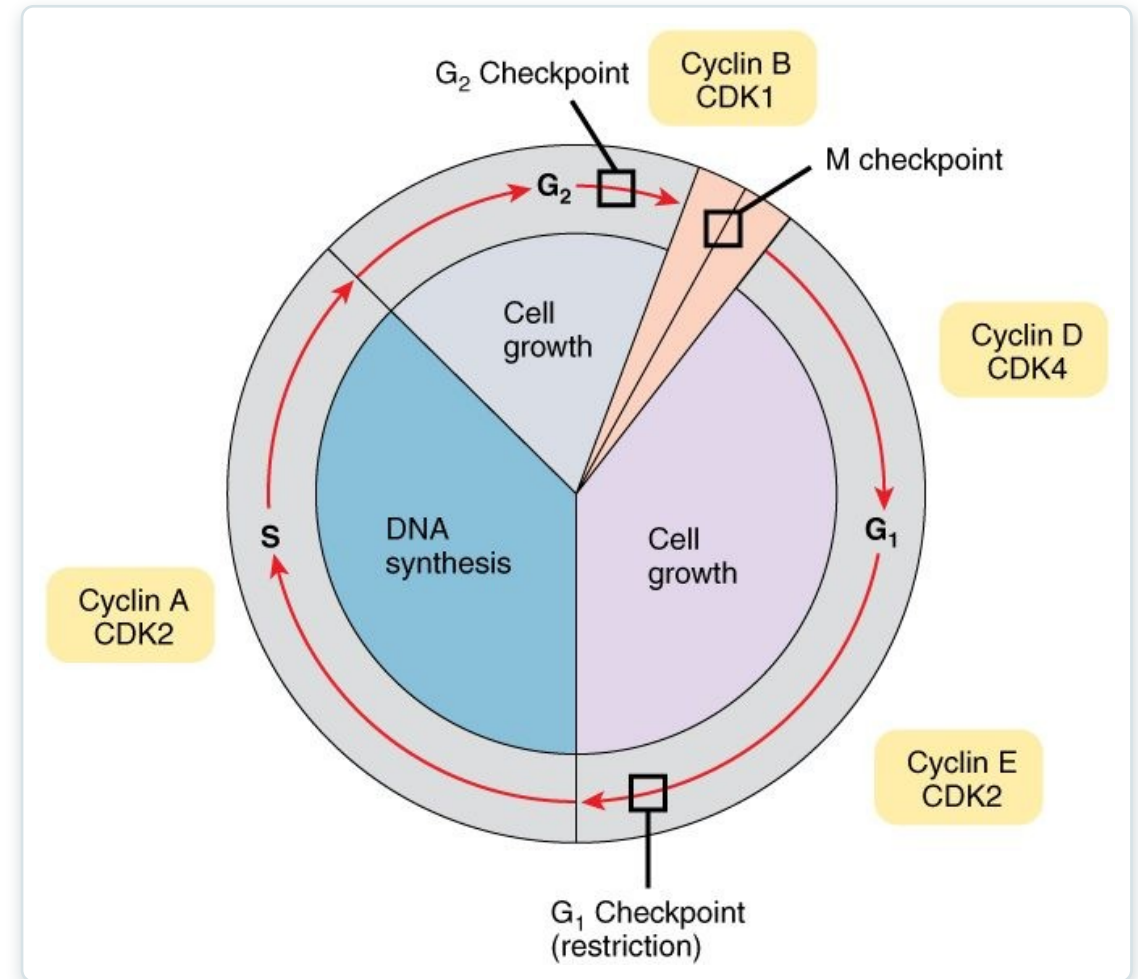
Cell Growth and Division

The cell cycle and mitosis — how one cell becomes two.

SECTION 3.5

The Cell Cycle

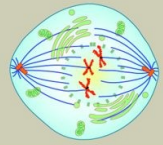
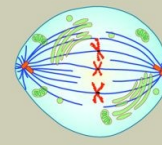
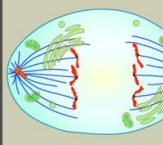
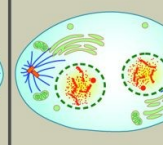
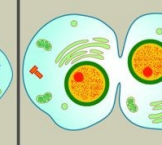
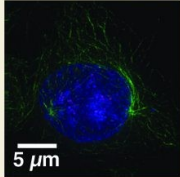
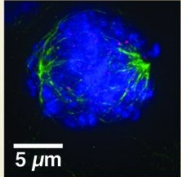
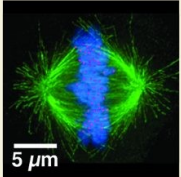
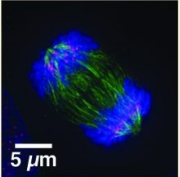
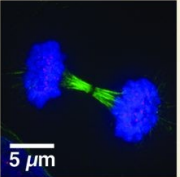
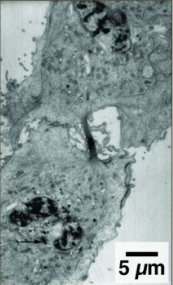
- Interphase: G₁, S, G₂ (growth + DNA copy)
- Mitotic phase: cell division
- Checkpoints control progression
- Cyclins regulate the cycle



The cell cycle and its checkpoints

Mitosis

- Prophase → metaphase → anaphase → telophase
- Chromosomes separate to opposite poles
- Cytokinesis splits the cytoplasm
- Two identical daughter cells

Prophase	Prometaphase	Metaphase	Anaphase	Telophase	Cytokinesis
					
• Chromosomes condense and become visible • Spindle fibers emerge from the centrosomes • Nuclear envelope breaks down • Centrosomes move toward opposite poles	• Chromosomes continue to condense • Kinetochores appear at the centromeres • Mitotic spindle microtubules attach to kinetochores	• Chromosomes are lined up at the metaphase plate • Each sister chromatid is attached to a spindle fiber originating from opposite poles	• Centromeres split in two • Sister chromatids (now called chromosomes) are pulled toward opposite poles • Certain spindle fibers begin to elongate the cell	• Chromosomes arrive at opposite poles and begin to decondense • Nuclear envelope material surrounds each set of chromosomes • The mitotic spindle breaks down • Spindle fibers continue to push poles apart	• Animal cells: a cleavage furrow separates the daughter cells • Plant cells: a cell plate, the precursor to a new cell wall, separates the daughter cells
					
MITOSIS					

The stages of mitosis



3.6

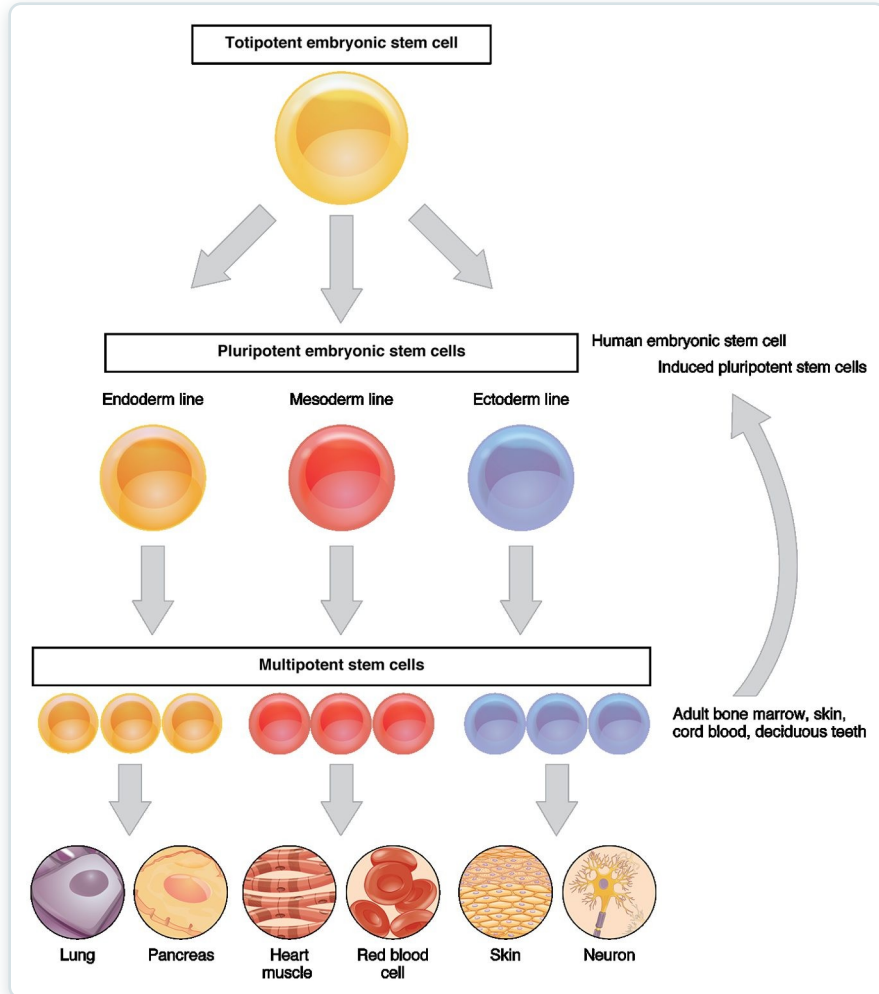
SECTION

Cellular Differentiation

How one fertilized egg gives rise to hundreds of specialized cell types.

SECTION 3.6

Stem Cells & Differentiation



Stem cell potency and differentiation

- Stem cells can become many cell types
- Totipotent → pluripotent → multipotent
- Differentiation specializes cells
- Basis of growth, repair, and therapy

Key Takeaways

- The cell membrane is a selectively permeable bilayer that controls transport.
- Organelles divide the labor: ER and Golgi process proteins, mitochondria make ATP.
- The nucleus stores DNA and directs the cell; DNA replicates before division.
- Protein synthesis flows from DNA to RNA to protein.
- The regulated cell cycle and mitosis drive growth, repair, and — when uncontrolled — cancer.

CLINICAL CONNECTIONS

Why the Cell Matters in Practical Nursing

Fluid & Electrolytes

Membrane transport, especially the Na^+/K^+ pump, underlies all fluid balance.

Immune Defense

Phagocytosis is how white blood cells engulf and destroy pathogens.

Energy & Fatigue

Mitochondrial ATP production fuels every active process in the body.

Cancer

Failed cell-cycle checkpoints cause uncontrolled cell division.

Medications

Many drugs act on membrane transport, enzymes, or protein synthesis.

Tissue Repair

Mitosis and stem cells drive healing after injury.